

Voluntary Report – Voluntary - Public Distribution

Date: June 18, 2026

Report Number: TW2026-0022

Report Name: Taiwan Revises National Standards for Automotive Unleaded Gasoline and Denatured Fuel Ethanol

Country: Taiwan

Post: Taipei

Report Category: Biofuels

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Report Highlights:

On June 9, 2026, Taiwan's Bureau of Standards, Metrology and Inspection (BSMI) published revised national standards for automotive unleaded gasoline (CNS 12614:2026) and denatured fuel ethanol (CNS 15109:2026). Both standards are modeled on ASTM D4814-25 and ASTM D4806-25 respectively but are not equivalent to those standards. Taiwan has incorporated stricter limits and additional parameters reflecting local environmental regulations and European fuel quality benchmarks. Both standards are voluntary under Taiwan's Standards Act unless incorporated into binding regulations by the relevant competent authority.

Background and Voluntary Status

BSMI periodically revises its Chinese National Standards (CNS) to reflect updates in international reference standards. CNS 12614:2026 replaces CNS 12614:2020, and CNS 15109:2026 replaces CNS 15109:2007.

Both standards are implemented on a voluntary basis. As stated in the prefatory text of both CNS documents: "Pursuant to Article 4 of the Standards Act, national standards are implemented on a voluntary basis. However, where the relevant competent authority incorporates all or part of the content into regulations, those regulations shall govern."

Important Note: These standards are based on ASTM D4814-25 and ASTM D4806-25, but they are not equivalent. Taiwan has made deliberate modifications. Compliance with the underlying ASTM standard does not guarantee compliance with the CNS standard. Interested parties should obtain the full CNS standards directly from BSMI.

Key Differences from ASTM Base Standards

The table below summarizes the most significant points of difference. It is not exhaustive; readers requiring full compliance details should consult the CNS standards directly.

CNS 12614:2026 vs. ASTM D4814-25 (Automotive Unleaded Gasoline)

Parameter	ASTM D4814-25	CNS 12614:2026	Note
Octane rating method	AKI [(RON+MON)/2]	RON only	Different measurement basis
Sulfur content	~80 mg/kg	≤ 10.0 mg/kg	Stricter; per Taiwan environmental regulation
Benzene content	≤ 0.9 vol%	≤ 0.8 vol%	Stricter; per Taiwan environmental regulation
Aromatics content	Not specified	≤ 35.0 vol%	Added from EN 228
Olefin content	Not specified	≤ 18.0 vol%	Added from EN 228
Vapor pressure (no ethanol)	≤ 62 kPa	45–60 kPa	Narrower range
Seasonal grades	7 vapor pressure/distillation classes	Not applicable	Taiwan is a single climate zone

Note: The ASTM values above are drawn from the CNS/ASTM comparison tables published within CNS 12614:2026. This table is a simplified summary. Parameters, units, and test methods may differ between the two standards in ways not captured here.

CNS 15109:2026 vs. ASTM D4806-25 (Denatured Fuel Ethanol)

Parameter	ASTM D4806-25	CNS 15109:2026	Note
Ethanol content (min)	92.1 vol%	94.5 vol%	Stricter; carried forward from CNS 15109:2007. CNS 15109:2026 specifies that ethanol content is measured after denaturant addition, with denaturant capped at 5 vol%.
Sulfur content	≤ 30.0 mg/kg	≤ 10.0 mg/kg	Stricter; per Taiwan environmental regulation
Electrical conductivity	Not specified	≤ 500 μS/m	Retained from CNS 15109:2007
Appearance/particulates	Not specified	Required	Added with reference to EN 15376

Note: *The ASTM values above are drawn from the CNS/ASTM comparison tables published within CNS 15109:2026. This table is a simplified summary. Parameters, units, and test methods may differ between the two standards in ways not captured here.*

Standard Availability

Full texts may be purchased from:

- **CNS Online:** <https://www.cnsonline.com.tw>
[CNS 12614 K5140](#)
[CNS 15109 K5157](#)

Attachments:

No Attachments.